

PRACTICAL TASK 1

~~~1. Gender Wage Gap in 2025~~~

Which three countries have the lowest gender wage gap?

Costa Rica, Belgium, and Denmark.

Which three countries have the highest gender wage gap?

Chile, Japan, and Korea.

Do some research on the country with the lowest gender wage gap and comment on why you think it succeeded in achieving a low gender wage gap in 2015 (max. 150 words).

Costa Rica has achieved the lowest gender wage gap due to proper and extensive gender equity policies, childcare support, and constant efforts to close the wage gap (Callen, 2016). This results in a smaller wage gap between genders.

~~~2. Isopropanol Sales Analysis~~~

Explain what is happening in the graph during March 2020 with regard to isopropanol sales (max. 100 words).

When viewing the isopropanol sales in March 2020, a significant increase can be observed when compared to previous month's sales, possibly related to the pandemic known as COVID-19 which increased global demand for isopropanol during that time period.

Describe a possible reason for the observation you made about isopropanol sales in March 2020 (max. 100 words).

As seen in on the graph, there was a big spike in isopropanol sales in March 2020. This sudden increase in sales was due to the COVID-19 pandemic, during which isopropanol-containing products such as hand sanitisers were in high demand(Koray, 2020).

~~~3. CO2 Emissions vs. GDP~~~

Discuss the relationship between CO2 emissions per person and GDP per capita for each continent listed in the figure legend (max. 350 words).

North America: Countries with high GDP per capita also exhibit high CO2 emissions per person. This showcases how industrialised economies have a high energy consumption, consequently releasing more CO2.

Europe: There is a moderate correlation between GDP and CO2 emissions. CO2 emissions is not as high as expected for a high GDP per capita country, and could be a result of sustainable practices and green energy policies.

Asia: There are different correlations and trends within this continent. High-GDP countries, such as Japan, has moderate emissions, while higher populated nations, such as china China, show high emissions due energy demands.

Africa: Most African countries have low GDP per capita and low emissions, which could be due to a lower energy demand and smaller economy.

South America: A mild correlation exists between GDP and emissions. Countries with higher GDPs tend to have higher emissions but are generally lower compared to other high gdp continents, such as North America

PRACTICAL TASK 2

~~~Scatterplot Matrix~~~

What do the graphs along the diagonal represent?

The diagonal axis represent histograms, showing the distribution of each individual variable, such as sale price, year built, garage size, and living area size.

Are most garages in Ames larger or smaller than 1000 ft²?

Most garages are smaller than 1000 ft².

Are the most expensive houses in Ames built before or after 1950?

After 1950

Describe the relationship between ‘Size of Living Area’ and ‘Sale Price.’

There is a strong correlation between the size of the living area and the sale price. The larger the living areas, the higher the sale price

REFERENCES

Callen, T., 2016. Purchasing Power Parity: Weights Matter. International Monetary Fund (IMF). Available at: <https://www.imf.org/en/Publications/fandd/issues/Series/Back-to-Basics/Purchasing-Power-Parity-PPP> [Accessed 17 January 2025].

Koray, D., 2020. US IPA prices rise on unprecedented hand sanitizer demand. ICIS. Available at: <https://www.icis.com/explore/resources/news/2020/03/26/10487220/us-ipa-prices-rise-on-unprecedented-hand-sanitizer-demand/> [Accessed 17 January 2025].